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Handwave lends a hand
to retailers with its
European alternative to
Amazon's palm
payments



Paying with a handwave once sounded like science fiction, but [contactless palm recognition service Amazon One](#) has already been used [more than 8 million times](#), according to the company. That's Amazon, though, which explains why it has been deployed in Amazon stores and more than 500 Whole Foods Market stores in the U.S., but only 150 third-party locations.

Meanwhile, fintech startups like Latvia's [Handwave](#) are stepping onto the field, aiming to provide third-party retailers with a similar but independent solution for faster checkout while leveraging the giant's role in popularizing biometric payments in the West. (China has already begun adopting biometric palm payments, with Tencent working to [bring its Weixin Palm Pay service into mainstream use](#).)

Like Apple's Face ID, palm scanning uses more than static images: It analyzes palm vein patterns and also verifies that the user is physically present when they hover their hand over the scanner. This method works for secure contactless payments and also applies to broader identity verification scenarios — with players like Keyo also supporting [secure building access and other applications](#).

In contrast, Handwave is focusing specifically on retail — and since it doesn't own stores like Amazon, it had to seek partners, which required having a product. Three years in, and now with its

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own hardware and software, the Latvian startup is preparing for market pilots that will deploy its palm-scanning devices at retail stores.

Merchants who deploy the startup's tech would pay a transaction fee that Handwave claims will be on par with or lower than standard payments.

According to Handwave, faster and cheaper checkouts could reduce costs. But unlike some cost-cutting measures, this solution aims to make things easier for customers — with the promise of no cards, no apps, no fingerprint scanners, and no facial scans — even for age verification and loyalty programs.

Handwave's co-founders, CEO Janis Stirna and Sandis Osmanis-Usmanis, previously worked for one of the world's largest global payment providers, Worldline. Despite this connection, the team aims to build a wide ecosystem. "Our plan is to collaborate with any financial institution or acquiring bank," Stirna told TechCrunch.

The startup has only partnered with a handful of financial institutions so far, "but very big ones, especially in Europe," Stirna said. This summer, the startup signed an agreement with Visa that could speed up the deployment of Handwave's solution in any country, according to its chief revenue officer, Oskars Laksevics.

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While Handwave has its eyes on the U.S. market as well, Laksevics believes that it can be an advantage to start out in the European Union — “the strictest market in the world” — and demonstrate compliance there before expanding.

Being an independent European player could also help the startup keep an edge if or when Amazon decides to more aggressively offer Amazon One to third parties; or if JP Morgan rolls out [its own palm payment experiment](#) further.

The startup also can rely on other arguments, including pricing. After financial partners told Handwave its devices need to be able to compete on price, the startup developed its own hardware and algorithms making them cheaper than others, Stirna said.

Being based in Riga also enabled Handwave to operate with limited capital. The startup told TechCrunch its R&D process was funded through bootstrapping, a \$780,000 angel investment round, and \$267,000 in non-equity funding. This sum came from an EU-funded [cybersecurity grant](#), as well as support from Latvia’s [LIAA Business Incubator](#) and EU-backed accelerator [Ready2Scale](#).

As it gears up for its first pilots and obtaining regulatory certifications, Handwave has now secured a \$4.2 million seed funding round led by Vilnius-based VC firm Practica Capital, with participation from FirstPick and Outlast Fund, also

from Lithuania; and Inovo.vc, a Polish VC firm that also operates in the Baltics.

The Baltic states have established themselves as a fintech hub, but also have scientific talent that’s easier for a startup like Handwave to attract and afford than in Silicon Valley — including AI engineers. “In the Baltics, there are not a lot of companies where you can get that extreme level of technical challenge to resolve,” Stirna said.

As for Laksevics, who previously held a senior marketing role at Baltic bank Luminor Bank, where Stirna had also worked, he told TechCrunch that he was drawn by the vision. “I left a very well-paid corporate job to join this one, and I truly believe that we are building the next big global payment platform,” he said.

Handwave seems ready to put its best hand forward — but only time will tell if the market will grab on and if biometric palm payments will truly take hold.

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Anna Heim

Freelance Reporter | [X](#) [in](#)

Anna Heim is a writer and editorial consultant.

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Amazon expands palm-scanning payment tech to 65 more Whole Foods locations

Lauren Forristal — 9:58 AM PDT · August 10, 2022

Amazon's "One" palm scanner payment technology will be launching at over 65 Whole Foods stores in California. This is the biggest rollout to date, with stores in Malibu, Montana Avenue, Santa Monica, Los Angeles, Orange County, Sacramento, the San Francisco Bay Area, and Santa Cruz receiving the tech that aims to modernize retail shopping.

When the checkout devices were [first announced](#) in 2020, Amazon One was available in Amazon Go stores, with an eventual expansion to [Whole Foods stores](#) in Austin, Seattle, Los Angeles, and New York City. Customers can also try checking out with their

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hand at the Amazon Style fashion store in Glendale, California, along with select Fresh and Go stores.

Amazon One is part of the company's mission to use "contactless" technology that makes it faster to pay. The tech works like this: Users visit a kiosk or a point-of-sale station at participating locations to link their palm and payment card to the service. Then, all they have to do during the checkout process is hover their hand over a scanner to complete the transaction.

Amazon One creates palm signatures using machine learning to identify customers. While the kiosk takes a picture of a user's palm, the company says it doesn't store the image there but instead encrypts it and sends it to a server for matching.

As Amazon customers continue giving up their data for a more convenient shopping experience, concerns increase about privacy.

If you use Face ID or fingerprint scanners, you already use [biometric data](#). However, it's likely some users aren't okay with the idea that Amazon One may allow the company to track your movements.

Even a [group of U.S. senators](#) expressed their concerns about the palm-scanning system. Senators Amy Klobuchar, Bill Cassidy and Jon Ossoff wrote in an open letter to Amazon chief executive Andy Jassy, "In contrast with biometric systems like Apple's Face ID and Touch ID or Samsung Pass, which store biometric information on a user's device, Amazon One reportedly uploads biometric information to the cloud, raising unique security risks."

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Last year, [Amazon partnered with ticketing company AXS](#), with plans to implement Amazon One at Denver, Colorado's Red Rocks Amphitheatre. Shortly after the announcement, hundreds of music fans, artists and human rights groups demanded Red Rocks drop the technology and ban all biometric surveillance tools like palm scans and facial recognition. They even signed a [letter](#) citing concerns about Amazon sharing palm print data with government agencies and potential hackers stealing the data from the cloud.

People aren't wrong to be bothered by it, as Amazon has been known to [store Alexa voice data](#). Also, Amazon [sold biometric facial recognition services](#) to United States law enforcement, and its Ring camera company [works with police](#) as well.

The company may have also found more data collection methods as it [acquired iRobot](#) last week, the [smart](#) room-mapping robot vacuum cleaner laden with advanced sensors.

Amazon expands its biometric-based Amazon One palm reader system to more retail stores



Last fall, Amazon introduced a new biometric device, Amazon One, that allowed customers to pay at Amazon Go stores using their palm. Today, the company says the device is being rolled out

to additional Amazon stores in Seattle — an expansion that will make the system available across eight total Amazon physical retail stores, including [Continue reading](#)

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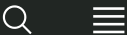
Lauren covers media, streaming, apps and platforms at TechCrunch.

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Lawmakers ask Amazon what it plans to do with palm print biometric data

Taylor Hatmaker — 2:00 AM PDT · August 13, 2021

A group of senators sent new Amazon CEO Andy Jassy a letter Friday pressing the company for more information about how it scans and stores customer palm prints for use in some of its retail stores.

IMAGE CREDITS:

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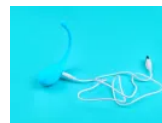
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The company rolled out the palm print scanners through a program it calls [Amazon One](#), encouraging people to make contactless payments in its brick and mortar stores without the use of a card. Amazon [introduced its Amazon One scanners late last year](#), and they can now be found in Amazon Go convenience and grocery stores, Amazon Books and Amazon four-star stores across the U.S. The scanners are also installed in eight Washington state-based [Whole Foods locations](#).

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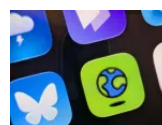
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Amazon expands its biometric-based Amazon One palm reader system to more retail stores



Last fall, Amazon introduced a new biometric device, Amazon One, that allowed customers to pay at Amazon Go stores using their palm. Today, the company says the device is being rolled out

to additional Amazon stores in Seattle — an expansion that will make the system available across eight total Amazon physical retail stores, including [Continue reading](#)

In the new letter, Senators Amy Klobuchar (D-MN), Bill Cassidy (R-LA) and Jon Ossoff (D-GA) press Jassy for details about how Amazon plans to expand its biometric payment system and if the data collected will help the company target ads.

“Amazon’s expansion of biometric data collection through Amazon One raises serious questions about Amazon’s plans for this data and its respect for user privacy, including about how Amazon may use the data for advertising and tracking purposes,” the senators wrote in the letter, embedded below.

The lawmakers also requested information on how many people have enrolled in Amazon One to date, how Amazon will secure the sensitive data and if the company has ever paired the palm prints with facial recognition data it collects elsewhere.



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“In contrast with biometric systems like Apple’s Face ID and Touch ID or Samsung Pass, which store biometric information on a user’s device, Amazon One reportedly uploads biometric information to the cloud, raising unique security risks,” the senators wrote. “... Data security is particularly important when it comes to immutable customer data, like palm prints.”

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The company controversially introduced a [\\$10 credit for new users](#) who enroll their palm prints in the program, prompting an outcry from privacy advocates who see it as a cheap tactic to coerce people to hand over sensitive personal data.

There’s plenty of reason to be skeptical. Amazon has faced fierce criticism for its other big biometric data project, the [AI facial recognition software known as Rekognition](#), which the company provided to U.S. law enforcement agencies before eventually backtracking with a [moratorium on policing applications](#) for the software last year.

Amazon will pay you \$10 in credit for your palm print biometrics



How much is your palm print worth? If you ask Amazon, it's about \$10 in promotional credit if you enroll your palm prints in its checkout-free stores and link it to your Amazon account. Last year,

Amazon introduced its new biometric palm print scanners, Amazon One, so customers can pay for goods in some stores ... [Continue reading](#)

Amazon shareholders want the company to stop selling facial recognition to law enforcement



Amazon shareholders are demanding the company stop selling Rekognition, the company's facial recognition software, to law enforcement. Unless the board of directors determines the

technology "does not cause or contribute to actual or potential violations of civil and human rights," shareholders want Amazon to stop selling the software to government agencies. Rekognition, which is part ... [Continue reading](#)

Amazon's facial recognition moratorium has major loopholes



In a surprise blog post, Amazon said it will put the brakes on providing its facial recognition technology to police for one year, but refuses to say if the move applies to federal law enforcement agencies. The moratorium comes two days after IBM said in a letter it was leaving the facial recognition market altogether. ... [Continue reading](#)

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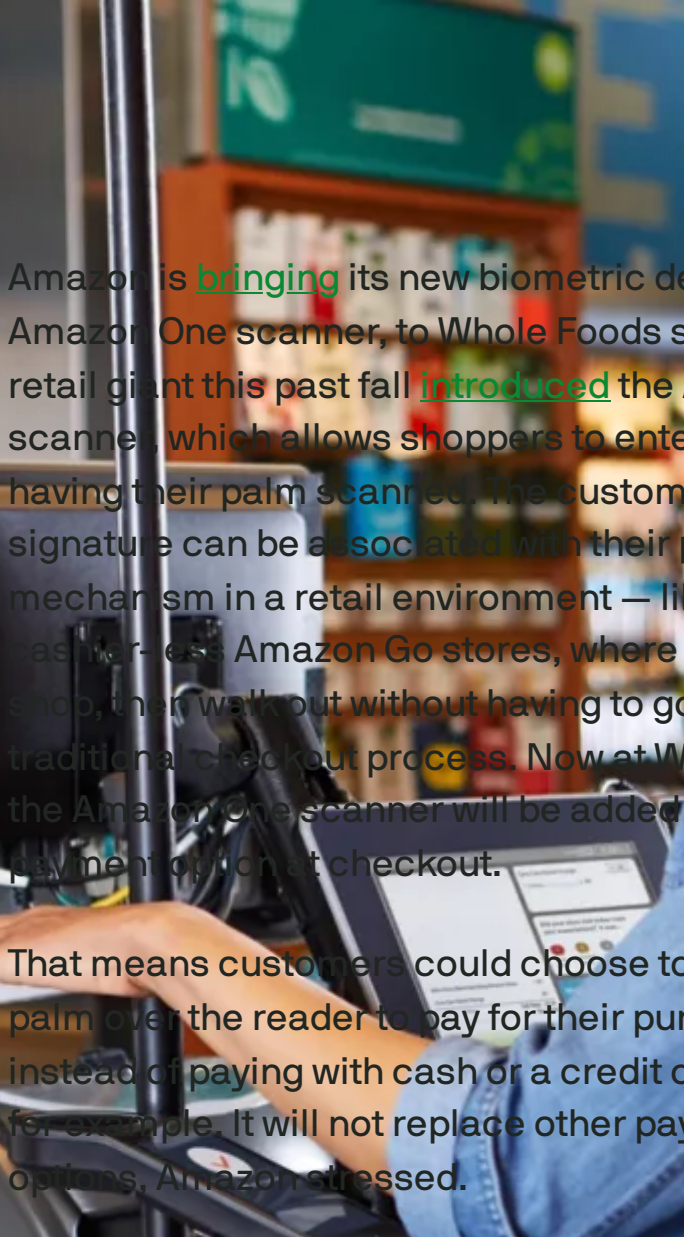
Taylor covered social media, gaming and culture at TechCrunch.

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Amazon is bringing its Amazon One palm scanner to select Whole Foods as a payment option



Amazon is [bringing](#) its new biometric device, the Amazon One scanner, to Whole Foods store. The retail giant this past fall [introduced](#) the Amazon One scanner, which allows shoppers to enter a store by having their palm scanned. The customer's palm signature can be associated with their payment mechanism in a retail environment — like Amazon's cashier-less Amazon Go stores, where customers shop, then walk out without having to go through a traditional checkout process. Now at Whole Foods, the Amazon One scanner will be added as a payment option at checkout.

That means customers could choose to scan their palm over the reader to pay for their purchases, instead of paying with cash or a credit or debit card, for example. It will not replace other payment options, Amazon stressed.

Amazon says it's initially adding the Amazon One palm scanner to the Whole Foods Market store at Madison Broadway in Seattle, but plans to roll it out to seven more Whole Foods Market stores in the greater Seattle area in the months to come. Seattle will likely serve as a test market for the new technology before Amazon chooses to roll it out more broadly.

Since [its launch in September](#), Amazon One scanners have already been installed in several Amazon stores in the Seattle area, including Amazon Go, Amazon Go Grocery, Amazon Books, Amazon 4-star and Amazon Pop Up. The retailer says “thousands” of customers have now enrolled their palm signatures.

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With the Whole Foods launch, customers will be able to sign up for Amazon One at a kiosk or device in the participating Whole Foods stores, where they can choose to enroll one or both palms. The scanner uses computer vision technology to create the unique palm signature, which is associated with the payment card the customer inserts into the device.

Existing customers who previously enrolled in Amazon One at a different location will have to re-insert their credit card one time in the Whole Foods store to continue to use the service in those stores, Amazon said.

In addition, customers will be able to link their Amazon One ID with their Amazon account in order to get their Prime membership discount to apply to their Whole Foods Market purchases via the Amazon One device in the future.

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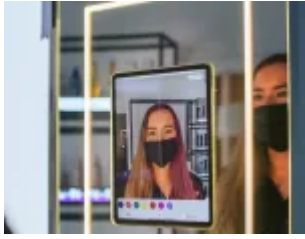
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“At Whole Foods Market, we’re always looking for new and innovative ways to improve the shopping experience for our customers,” said Arun Rajan, senior vice president of technology and chief technology officer at Whole Foods Market, in a statement about the expansion. “Working closely with Amazon, we’ve brought benefits like Prime member discounts, online grocery delivery and pickup, and free returns to our customers, and we’re excited to add Amazon One as a payment option beginning today. We’re starting with an initial store at Madison Broadway in Seattle and look forward to hearing what customers think as we expand this option to additional stores over time,” he added.

The device, of course, comes with concerns, given Amazon’s track record with biometric technology. The company has [sold biometric facial recognition services](#) to law enforcement in the U.S.; its facial recognition technology [was the subject of a data privacy lawsuit](#); and its Ring camera [company continues to work in partnership with police](#). It was also found to be [keeping Alexa voice records indefinitely](#), with no option for customers to delete them. (Amazon later [changed](#) that — and, to be fair, [Google](#) and [Apple](#) were mishandling customer voice data, too.)

Amazon is opening a London hair salon to test AR and other retail technologies



Amazon announced this morning it's opening Amazon Salon, the retailer's first hair salon and a place where Amazon aims to test new technologies with the general public. The salon will occupy over 1,500 sq. ft on Brushfield Street in

London's Spitalfields, where Amazon says it will initially be trialing the use of augmented reality (AR) [Continue reading](#)

Just yesterday, Amazon [announced tests](#) of other retail and AR technology in a London area hair salon, which would involve cameras capturing customer images. The company said it wasn't retaining "customer data," but declined to answer a question about the non-personal data being collected at the salon.

In today's announcement, Amazon notes that the Amazon One device is "protected by multiple security controls, and palm images are never stored on the Amazon One device." It says the images are encrypted and sent to a secure area built for Amazon One in the cloud where Amazon creates the customers' palm signatures. It also offers a way for customers to unenroll from Amazon One from either a device itself or from one.amazon.com, which also deletes their biometric data when all their transactions have completed.

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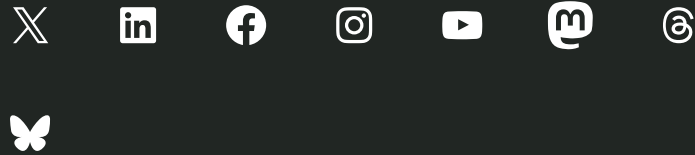
Sarah Perez

Consumer News Editor | X Butterfly @ in

Sarah has worked as a reporter for TechCrunch since August 2011. She joined the company after having...

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